

Healthcare Data Cleaning & Understanding

Healthcare Dataset Analysis Report

This practical demonstrates healthcare data exploration, identification of column types, cleaning of missing/duplicate records, and basic descriptive analysis. The dataset below is simulated for educational purposes and contains no real patient information.

1. Dataset Overview

The initial dataset contains 21 records and 6 columns. One duplicate patient record, one missing age value, and one missing disease value were identified.

Column	Type	Description
Patient ID	Integer	Unique identifier
Age	Numeric	Age in years
Gender	Categorical	Male/Female
Disease	Categorical	Health condition
Medication	Categorical	Medicine prescribed
Dosage	Text/Categorical	Dose strength and unit

2. Data Cleaning

- Removed the duplicate Patient ID 120 record.
- Filled the missing age using the median age of the valid records (44 years).
- Replaced the missing disease entry with "Not Reported" rather than inventing a diagnosis.
- Rechecked the cleaned dataset; 20 unique records remain.

3. Basic Analysis

Measure	Result
Records before cleaning	21
Records after cleaning	20
Duplicate records removed	1
Missing cells identified	2
Average age after cleaning	44.0 years
Most common disease	Diabetes & Hypertension (6 each)
Most common gender	Male (10) / Female (10)

4. Disease Distribution

Disease	Count
Diabetes	6
Hypertension	6
Asthma	3
Arthritis	2
Migraine	2
Not Reported	1

5. Cleaned Dataset

ID	Age	Gender	Disease	Medication	Dosage
101	22	Male	Diabetes	Metformin	500 mg
102	35	Female	Hypertension	Amlodipine	5 mg
103	47	Male	Diabetes	Metformin	500 mg
104	29	Female	Asthma	Salbutamol	100 mcg
105	61	Male	Hypertension	Losartan	50 mg
106	42	Female	Diabetes	Metformin	500 mg
107	55	Male	Arthritis	Ibuprofen	400 mg
108	31	Female	Migraine	Sumatriptan	50 mg
109	68	Male	Hypertension	Amlodipine	5 mg
110	26	Female	Asthma	Salbutamol	100 mcg
111	39	Male	Diabetes	Metformin	500 mg
112	50	Female	Hypertension	Losartan	50 mg
113	45	Male	Arthritis	Ibuprofen	400 mg
114	33	Female	Migraine	Sumatriptan	50 mg
115	72	Male	Hypertension	Amlodipine	5 mg
116	50	Female	Diabetes	Metformin	500 mg
117	28	Male	Not Reported	Salbutamol	100 mcg
118	58	Female	Hypertension	Losartan	50 mg
119	36	Male	Diabetes	Metformin	500 mg
120	44	Female	Asthma	Salbutamol	100 mcg

6. Conclusion

The healthcare dataset was successfully explored and cleaned. Duplicate data were removed, missing values were handled using a transparent method, and descriptive statistics were calculated. Diabetes and Hypertension were the most frequently recorded conditions, with 6 records each. The exercise demonstrates basic healthcare data-cleaning and understanding techniques.